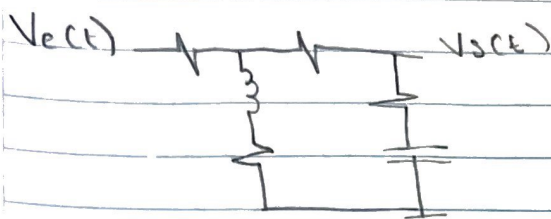


Munguía Silva Yesenia Isamar



Ecuaciones principales

$$V_c(t) = R i_1(t) + L \frac{d[i_1(t) + i_2(t)]}{dt} + R [i_1(t) - i_2(t)]$$

$$L \frac{d[i_1(t) - i_2(t)]}{dt} + R [i_1(t) - i_2(t)] = R i_2(t) + R i_2(t) + \frac{1}{C} \int i_2(t) dt$$

$$V_s(t) = R i_2(t) + \frac{1}{C} \int i_2(t) dt$$

Transformada de Laplace

$$V_c(s) = R i_1(s) + L s [i_1(s) - i_2(s)] + R [i_1(s) - i_2(s)]$$

$$L s [i_1(s) - i_2(s)] + R [i_1(s) - i_2(s)] = R i_2(s) + R i_2(s) + \frac{i_2(s)}{Cs}$$

$$V_s(s) = R i_2(s) + \frac{i_2(s)}{Cs} = \left(\frac{RCs + 1}{Cs} \right) i_2(s)$$

Procedimiento algebraico

$$V_c(s) = (Ls + 2R) i_1(s) - (Ls + R) i_2(s)$$

$$(Ls + R) i_1(s) = \left(\frac{CLs^2 + 3CRs + 1}{Cs} \right) i_2(s)$$

$$i_1(s) = \left(\frac{CLs^2 + 3CRs + 1}{Cs} \right) \left(\frac{1}{Ls + R} \right) i_2(s) = \left(\frac{CLs^2 + 3CRs + 1}{Cs(Ls + R)} \right) i_2(s)$$

$$\therefore V_c(s) = (Ls + 2R) \left(\frac{CLs^2 + 3CRs + 1}{Cs(Ls + R)} \right) i_2(s) - (Ls + R) i_2(s)$$

$$V_c(s) = \left(\frac{(Ls + 2R)(CLs^2 + 3CRs + 1) - Cs(Ls + R)^2}{Cs(Ls + R)} \right) i_2(s)$$

Función de transferencia

$$V_s(s) = \left(\frac{RCs + 1}{Cs} \right) i_2(s)$$

$$\frac{V_c(s)}{V_s(s)} = \frac{(Ls + 2R)(CLs^2 + 3CRs + 1) - Cs(Ls + R)^2}{Cs(Ls + R) \cdot (RCs + 1)}$$

$$= \frac{LCRs^2 + Ls + R^2Cs + R}{3LCRs^2 + 15R^2Cs + 15Ls + 2R}$$

$$= \frac{LCRs^2 + Ls + R^2Cs + R}{3LCRs^2 + 15R^2Cs + 15Ls + 2R}$$

$$= \frac{LCRs^2 + Ls + R^2Cs + R}{3LCRs^2 + 15R^2Cs + 15Ls + 2R}$$

$$= \frac{LCRs^2 + Ls + R^2Cs + R}{3LCRs^2 + 15R^2Cs + 15Ls + 2R}$$

Error en estado estacionario

$$e(s) = \lim_{s \rightarrow 0} v_e(s) \left(1 - \frac{v_s(s)}{v_e(s)} \right) = \lim_{s \rightarrow 0} \frac{1}{s} \left(\frac{1 - LCRs^2 + (L + R^2C)s + R}{3LCRs^2 + (CR^2 + L)s + 2R} \right) = 1 - \frac{1}{2} = 0.5V //$$

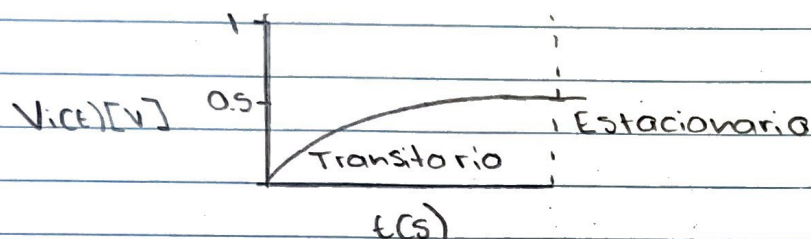
Polos del sistema y estabilidad en lazo abierto

$$C = 470 \times 10^{-6} F \quad a = 3LCR \quad \lambda_1 = -0.170$$

$$R = 5 \times 10^3 \Omega \quad b = 5R^2C + L \quad \lambda_2 = -8.33 \times 10^8$$

$$L = 10 \times 10^{-6} H \quad C = 2R$$

Sistema estable con respuesta sobreamortiguada



Modelo de ecuaciones integro-diferenciales

El modelo se expresa describiendo las variables dependientes, es decir, las corrientes:

$$v_e(t) = 2Ri_2(t) + L \frac{d[i_1(t) - i_2(t)]}{dt} - Ri_2(t)$$

$$i_2(t) = \left[v_e(t) + Ri_2(t) - L \frac{d[i_1(t) - i_2(t)]}{dt} \right] \frac{1}{2R}$$

$$L \frac{d[i_2(t) - i_1(t)]}{dt} + Ri_2(t) = Ri_2(t) + Ri_2(t) + Ri_2(t) + \frac{1}{C} \int i_2(t) dt$$

$$i_2(t) = \left[L \frac{d[i_2(t) - i_1(t)]}{dt} + Ri_2(t) - \frac{1}{C} \int i_2(t) dt \right] \frac{1}{3R}$$

$$v_s(t) = Ri_2(t) + \frac{1}{C} \int i_2(t) dt$$